

CALCULUS I

Chapter 3: INDEFINITE INTEGRALS

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INDEFINITE INTEGRALS

Definition 1 (Antiderivatives): A function $F(x)$ is called an **antiderivative** of $f(x)$ on an interval I if

$$F'(x) = f(x)$$

for all x in I .

Example 1: Let $f(x) = 2x$. If $F(x) = x^2$, then $F'(x) = 2x = f(x)$. But the function $G(x) = x^2 + 10$ also satisfies $G'(x) = 2x = f(x)$. Therefore both F and G are antiderivatives of f . Indeed, any function of the form $H(x) = x^2 + C$, where C is a constant, is an antiderivative of f . We have the following theorem.

Theorem 1: If F is an antiderivative of f on an interval I , then the most general antiderivative of f on I is $F(x) + C$ where C is an arbitrary constant.

Definition 2 (Indefinite Integrals): The functions of the type $F(x) + C$ where C is a constant are called the **indefinite integrals** of f on I , and denoted by

$$\int f(x) \, dx = F(x) + C$$

Properties of Indefinite Integrals

$$(1) \left(\int f(x) dx \right)' = f(x)$$

$$(2) \int Cf(x) dx = C \int f(x) dx$$

$$(3) \int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$$

Integrals of Fundamental Functions

$$(1) \int x^\alpha dx = \frac{x^{\alpha+1}}{\alpha+1} + C \quad (\alpha \neq -1), \quad \int \frac{dx}{x} = \ln|x| + C$$

$$(2) \int a^x dx = \frac{a^x}{\ln a} + C \quad (a > 0, a \neq 1), \quad \int e^x dx = e^x + C$$

$$(3) \int \cos x dx = \sin x + C, \quad \int \sin x dx = -\cos x + C$$

Integrals of Fundamental Functions

$$(4) \int \frac{dx}{\cos^2 x} = \tan x + C, \quad \int \frac{dx}{\sin^2 x} = -\cot x + C$$

$$(5) \int \cosh x \, dx = \sinh x + C, \quad \int \sinh x \, dx = \cosh x + C$$

$$(6) \int \frac{dx}{\cosh^2 x} = \tanh x + C, \quad \int \frac{dx}{\sinh^2 x} = -\coth x + C$$

$$(7) \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \arctan \frac{x}{a} + C \quad (a \neq 0)$$

$$(8) \int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C \quad (a \neq 0)$$

$$(9) \int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin \frac{x}{a} + C \quad (a > 0)$$

$$(10) \int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln \left| x + \sqrt{x^2 \pm a^2} \right| + C \quad (a \neq 0)$$

Example 2: Use the above formulas to find:

$$\begin{aligned}(1) \int (x^2 + 2x + 3) dx &= \int x^2 dx + 2 \int x dx + 3 \int dx \\ &= \frac{x^3}{3} + x^2 + 3x + C\end{aligned}$$

$$\begin{aligned}(2) \int \left(\frac{1}{x} - \frac{1}{x^2} \right) dx &= \int \frac{dx}{x} - \int x^{-2} dx \\ &= \ln |x| - \frac{x^{-2+1}}{-2+1} + C \\ &= \ln |x| + \frac{1}{x} + C\end{aligned}$$

$$\begin{aligned}
 (3) \quad \int \tan^2 x \, dx &= \int [(\tan^2 x + 1) - 1] \, dx \\
 &= \int \left(\frac{1}{\cos^2 x} - 1 \right) dx = \tan x - x + C
 \end{aligned}$$

$$\begin{aligned}
 (4) \quad \int \frac{dx}{\sin^2 x \cos^2 x} &= \int \frac{\sin^2 x + \cos^2 x}{\sin^2 x \cos^2 x} dx \\
 &= \int \frac{dx}{\cos^2 x} + \int \frac{dx}{\sin^2 x} \\
 &= \tan x - \cot x + C
 \end{aligned}$$

$$(5) \quad \int \frac{dx}{x^2 + 2} = \int \frac{dx}{x^2 + (\sqrt{2})^2} = \frac{1}{\sqrt{2}} \arctan \frac{x}{\sqrt{2}} + C$$

THE SUBSTITUTION RULE

The Substitution Rule

If $u = g(x)$ is a differentiable function whose range is an interval I and f is continuous on I , then

$$\int f(g(x)) \cdot g'(x) dx = \int f(u) du \quad (u = g(x))$$

Example 3:

$$\begin{aligned} (1) \int \frac{\cos x}{\sin^2 x} dx &= \left| \begin{array}{l} u = \sin x \\ du = \cos x dx \end{array} \right| = \int \frac{du}{u^2} = -\frac{1}{u} + C \\ &= -\frac{1}{\sin x} + C \end{aligned}$$

$$\begin{aligned} (2) \int \frac{x dx}{1+x^2} &= \left| \begin{array}{l} u = 1+x^2, du = 2x dx \\ \Rightarrow x dx = \frac{du}{2} \end{array} \right| = \frac{1}{2} \int \frac{du}{u} \\ &= \frac{1}{2} \ln |u| + C = \frac{1}{2} \ln(1+x^2) + C \end{aligned}$$

$$\begin{aligned}
 (3) \quad \int \sqrt{2x+1} \, dx &= \left| \begin{array}{l} u = 2x + 1, \, du = 2dx \\ dx = \frac{1}{2} du \end{array} \right| \\
 &= \int \sqrt{u} \frac{du}{2} = \frac{1}{2} \cdot \frac{u^{3/2}}{3/2} + C \\
 &= \frac{1}{3} (2x+1)^{3/2} + C
 \end{aligned}$$

$$\begin{aligned}
 (4) \quad \int x^3 \cos(x^4 + 2) \, dx &= \left| \begin{array}{l} u = x^4 + 2, \, du = 4x^3 dx \\ x^3 dx = \frac{1}{4} du \end{array} \right| \\
 &= \frac{1}{4} \int \cos u \, du = \frac{1}{4} \sin u + C \\
 &= \frac{1}{4} \sin(x^4 + 2) + C
 \end{aligned}$$

Using the Substitution Rule we have the following useful formulas:

- $\int \frac{u'(x)}{2\sqrt{u(x)}} dx = \sqrt{u(x)} + C, \quad (u(x) \geq 0)$
- $\int \frac{u'(x)}{u(x)} dx = \ln |u(x)| + C, \quad (u(x) \neq 0)$
- $\int \tan x \, dx = -\ln |\cos x| + C$
- $\int \cot x \, dx = \ln |\sin x| + C$
- $\int \frac{dx}{\cos x} = \frac{1}{2} \ln \left| \frac{1 + \sin x}{1 - \sin x} \right| + C = \ln \left| \frac{1 + \tan \frac{x}{2}}{1 - \tan \frac{x}{2}} \right| + C$
- $\int \frac{dx}{\sin x} = \frac{1}{2} \ln \left| \frac{1 - \cos x}{1 + \cos x} \right| + C = \ln \left| \tan \frac{x}{2} \right| + C$

INTEGRATION BY PARTS

Integration By Parts

If f and g are differentiable functions, then

$$\int f(x) \cdot g'(x) \cdot dx = f(x) \cdot g(x) - \int f'(x) \cdot g(x) \cdot dx$$

This formula is called the **formula for integration by parts**.

It is perhaps easier to remember in the following notation.

Let $u = f(x)$ and $v = g(x)$. Then the differentials are $du = f'(x)dx$ and $dv = g'(x)dx$, so, by the Substitution Rule, the formula for integration by parts becomes

$$\int u \cdot dv = u \cdot v - \int v \cdot du$$

Example 4:

$$\begin{aligned}(1) \int \ln x \, dx &= \left| \begin{array}{l} u = \ln x, dv = dx \\ \Rightarrow du = \frac{dx}{x}, v = x \end{array} \right| \\ &= x \ln x - \int x \frac{dx}{x} = x \ln x - x + C\end{aligned}$$

$$\begin{aligned}(2) \int x \cos x \, dx &= \left| \begin{array}{l} u = x, dv = \cos x \, dx \\ du = dx, v = \sin x \end{array} \right| \\ &= x \sin x - \int \sin x \, dx \\ &= x \sin x + \cos x + C\end{aligned}$$

$$\begin{aligned}
 (3) \int \arctan x \, dx &= \left| \begin{array}{l} u = \arctan x, dv = dx \\ \Rightarrow du = \frac{dx}{1+x^2}, v = x \end{array} \right| \\
 &= x \arctan x - \int \frac{x}{1+x^2} \, dx = \\
 &= x \arctan x - \frac{1}{2} \int \frac{2x}{1+x^2} \, dx \\
 &= x \arctan x - \frac{1}{2} \ln(1+x^2) + C
 \end{aligned}$$

$$\begin{aligned}
 (4) \quad \int x^2 e^{2x} dx &= \left| \begin{array}{l} u = x^2, dv = e^{2x} dx \\ \Rightarrow du = 2x dx, v = \frac{1}{2} e^{2x} \end{array} \right| \\
 &= \frac{x^2}{2} e^{2x} - \int x e^{2x} dx = \\
 &= \left| \begin{array}{l} u = x, dv = e^{2x} dx \\ \Rightarrow du = dx, v = \frac{1}{2} e^{2x} \end{array} \right| \\
 &= \frac{x^2}{2} e^{2x} - \left(\frac{x}{2} e^{2x} - \int \frac{1}{2} e^{2x} dx \right) \\
 &= \frac{x^2}{2} e^{2x} - \frac{x}{2} e^{2x} + \frac{1}{4} e^{2x} + C \\
 &= \frac{1}{4} (2x^2 - 2x + 1) e^{2x} + C
 \end{aligned}$$

$$\begin{aligned}
 (5) \quad \int e^x \sin x \, dx &= \left| \begin{array}{l} u = e^x, dv = \sin x dx \\ \Rightarrow du = e^x dx, v = -\cos x \end{array} \right| \\
 &= -e^x \cos x + \int e^x \cos x \, dx = \\
 &= \left| \begin{array}{l} u = e^x, dv = \cos x dx \\ \Rightarrow du = e^x dx, v = \sin x \end{array} \right| \\
 &= -e^x \cos x + e^x \sin x - \int e^x \sin x \, dx
 \end{aligned}$$

From here we obtain

$$\int e^x \sin x \, dx = \frac{e^x}{2} (\sin x - \cos x) + C$$

Example 5: Evaluate

$$\mathcal{C} = \int e^{ax} \cos bx \, dx \quad \text{and} \quad \mathcal{S} = \int e^{ax} \sin bx \, dx$$

Solution : Let $\mathcal{Z} = \mathcal{C} + \mathbf{i}\mathcal{S}$ where $\mathbf{i}$ is a complex unit number ($\mathbf{i}^2 = -1$). We have

$$\begin{aligned}\mathcal{Z} &= \int e^{ax} \cos bx \, dx + \mathbf{i} \int e^{ax} \sin bx \, dx \\&= \int e^{ax} (\cos bx + \mathbf{i} \sin bx) \, dx = \int e^{ax} e^{\mathbf{i}bx} \, dx \\&= \int e^{(a+\mathbf{i}b)x} \, dx = \frac{1}{a + \mathbf{i}b} e^{(a+\mathbf{i}b)x} + C \\&= \frac{a - \mathbf{i}b}{a^2 + b^2} e^{ax} (\cos bx + \mathbf{i} \sin bx) + C\end{aligned}$$

We have $\mathcal{C} = \operatorname{Re}(\mathcal{Z})$, $\mathcal{S} = \operatorname{Im}(\mathcal{Z})$, and finally

$$(1) \int e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) + C$$

$$(2) \int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) + C$$

INTEGRATION OF RATIONAL FUNCTIONS

Integration Of Rational Functions

In this section we show how to integrate any **rational function** (a ratio of two polynomials) by expressing it as a sum of simpler fractions, called **partial fractions**, that we already know how to integrate. To illustrate the method, observe that by taking the fractions $\frac{2}{x-1}$ and $\frac{1}{x+2}$ to a common denominator we obtain

$$\frac{2}{x-1} - \frac{1}{x+2} = \frac{2(x+2) - (x-1)}{(x-1)(x+2)} = \frac{x+5}{x^2+x-2}$$

If we now reverse the procedure, we see how to integrate the function on the right side of this equation:

$$\begin{aligned}
 \int \frac{x+5}{x^2+x-2} dx &= \int \left(\frac{2}{x-1} - \frac{1}{x+2} \right) dx \\
 &= \int \frac{2}{x-1} dx - \int \frac{1}{x+2} dx \\
 &= 2 \ln |x-1| - \ln |x+2| + C
 \end{aligned}$$

To see how the method of partial fractions works in general, let's consider a rational function

$$f(x) = \frac{P(x)}{Q(x)}$$

where P and Q are polynomials. It's possible to express f as a sum of simpler fractions provided that the degree of P is less than the degree of Q . Such a rational function is called **proper**.

If f is **improper**, that is, $\deg(P) \geq \deg(Q)$, then we must take the preliminary step of dividing Q into P (by long division) until a remainder $R(x)$ is obtained such that $\deg(R) < \deg(Q)$. The division statement is

$$f(x) = \frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}$$

where S and R are also polynomials.

Example 6: Find $\int \frac{x^3 + x}{x - 1} dx$

Solution : Since the degree of the numerator is greater than the degree of the denominator, we first perform the long division.

$$\frac{x^3 + x}{x - 1} = \frac{(x - 1)(x^2 + x + 2) + 2}{x - 1} = x^2 + x + 2 + \frac{2}{x - 1}$$

This enables us to write

$$\begin{aligned} \int \frac{x^3 + x}{x - 1} dx &= \int \left(x^2 + x + 2 + \frac{2}{x - 1} \right) dx \\ &= \frac{x^3}{3} + \frac{x^2}{2} + 2x + 2 \ln |x - 1| + C \end{aligned}$$

The next step is to factor the denominator $Q(x)$ as far as possible. It can be shown that any polynomial Q can be factored as a product of **linear factors** (of the form $ax + b$) and **irreducible quadratic factors** (of the form $ax^2 + bx + c$, where $b^2 - 4ac < 0$). For instance, if $Q(x) = x^4 - 16$, we could factor it as

$$Q(x) = (x^2 - 4)(x^2 + 4) = (x - 2)(x + 2)(x^2 + 4)$$

The third step is to express the proper rational function $\frac{R(x)}{Q(x)}$ as a sum of partial fractions of the form

$$\frac{A}{(ax + b)^i} \quad \text{or} \quad \frac{Ax + B}{(ax^2 + bx + c)^j}$$

A theorem in algebra guarantees that it is always possible to do this. We explain the details for the four cases that occur.

CASE I: The denominator $Q(x)$ is a product of distinct linear factors.

This means that we can write

$$Q(x) = (a_1x + b_1)(a_2x + b_2) \dots (a_kx + b_k)$$

where no factor is repeated (and no factor is a constant multiple of another). In this case the partial fraction theorem states that there exist constants $A_1, A_2, \dots, A_k$ such that

$$\frac{R(x)}{Q(x)} = \frac{A_1}{a_1x + b_1} + \frac{A_2}{a_2x + b_2} + \dots + \frac{A_k}{a_kx + b_k}$$

These constants can be determined as in the following example.

Example 7: Evaluate $I = \int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx$

Solution : Since the degree of the numerator is less than the degree of the denominator, we don't need to divide. We factor the denominator as

$$2x^3 + 3x^2 - 2x = x(2x^2 + 3x - 2) = x(2x - 1)(x + 2)$$

Since the denominator has three distinct linear factors, the partial fraction decomposition of the integrand has the form

$$\frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} = \frac{A}{x} + \frac{B}{2x - 1} + \frac{C}{x + 2}$$

To determine the values of A , B , and C , we multiply both sides of this equation by the product of the denominators, $x(2x - 1)(x + 2)$, obtaining

$$x^2 + 2x - 1 = A(2x - 1)(x + 2) + Bx(x + 2) + Cx(2x - 1) \quad (1)$$

There are two methods for determining these coefficients:

Method 1: Expanding the right side of the equation (1) and writing it in the standard form for polynomials, we get

$$x^2 + 2x - 1 = (2A + B + 2C)x^2 + (3A + 2B - C)x - 2A$$

The polynomials in the above equation are identical, so their coefficients must be equal.

The coefficient of x^2 on the right side, $2A + B + 2C$, must equal the coefficient of x^2 on the left side, namely, 1. Likewise, the coefficients of x are equal and the constant terms are equal. This gives the following system of equations for A , B , and C :

$$\begin{cases} 2A + B + 2C = 1 \\ 3A + 2B - C = 2 \\ -2A = -1 \end{cases} \implies A = \frac{1}{2}, B = \frac{1}{5}, C = -\frac{1}{10}$$

Method 2: Equation (1) is an identity; it is true for every value of x . Let's choose values of x that simplify the equation. Typically, we choose x to be a root of the polynomial in the denominator.

- If we put $x = 0$ in Equation (1), then the second and third terms on the right side vanish and the equation then becomes $-2A = -1$, or $A = \frac{1}{2}$.
- Putting $x = \frac{1}{2}$ gives $\frac{5B}{4} = \frac{1}{4} \Rightarrow B = \frac{1}{5}$.
- Putting $x = -2$ gives $10C = -1 \Rightarrow C = -\frac{1}{10}$.

Finally,

$$\begin{aligned}
 I &= \int \left[\frac{1}{2} \cdot \frac{1}{x} + \frac{1}{5} \cdot \frac{1}{2x-1} - \frac{1}{10} \cdot \frac{1}{x+2} \right] dx \\
 &= \frac{1}{2} \ln|x| + \frac{1}{10} \ln|2x-1| - \frac{1}{10} \ln|x+2| + C \\
 &= \frac{1}{10} \ln \left| \frac{x^5(2x-1)}{x+2} \right| + C
 \end{aligned}$$

CASE II: The denominator $Q(x)$ is a product of linear factors, some of which are repeated.

Suppose the first linear factor $a_1x + b_1$ is repeated r times; that is, $(a_1x + b_1)^r$ occurs in the factorization of $Q(x)$. Then instead of the single term $\frac{A_1}{a_1x + b_1}$, we would use

$$\frac{A_1}{a_1x + b_1} + \frac{A_2}{(a_1x + b_1)^2} + \cdots + \frac{A_r}{(a_1x + b_1)^r}$$

We consider the following example.

Example 8: Evaluate $I = \int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx$

Solution : Since the degree of the numerator is greater than the degree of the denominator, then the first step is to divide.

The result is

$$\frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} = x + 1 + \frac{4x}{x^3 - x^2 - x + 1}$$

The second step is to factor the denominator. We obtain

$$x^3 - x^2 - x + 1 = (x - 1)(x^2 - 1) = (x - 1)^2(x + 1)$$

Since the linear factor $x - 1$ occurs twice, the partial fraction decomposition is

$$\frac{4x}{x^3 - x^2 - x + 1} = \frac{4x}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1}$$

Multiplying by the least common denominator, $(x-1)^2(x+1)$, we get

$$\begin{aligned} 4x &= A(x-1)(x+1) + B(x+1) + C(x-1)^2 \\ &= (A+C)x^2 + (B-2C)x + (-A+B+C) \end{aligned}$$

$$\Rightarrow \begin{cases} A + C = 0 \\ B - 2C = 4 \\ -A + B + C = 0 \end{cases} \Rightarrow A = 1, B = 2, C = -1$$

So

$$\begin{aligned} I &= \int \left[x + 1 + \frac{1}{x-1} + \frac{2}{(x-1)^2} - \frac{1}{x+1} \right] dx \\ &= \frac{x^2}{2} + x + \ln|x-1| - \frac{2}{x-1} - \ln|x+1| + C \\ &= \frac{x^2}{2} + x - \frac{2}{x-1} + \ln \left| \frac{x-1}{x+1} \right| + C \end{aligned}$$

CASE III: The denominator $Q(x)$ contains irreducible quadratic factors, none of which is repeated.

If $Q(x)$ has the factor $ax^2 + bx + c$, where $b^2 - 4ac < 0$, then the expression for $\frac{R(x)}{Q(x)}$ will have a term of the form

$$\frac{Ax + B}{ax^2 + bx + c}$$

where A and B are constants to be determined.

Example 9: Find $I = \int \frac{2x^2 - x + 4}{x^3 + 4x} dx$

Solution : Since $x^3 + 4x = x(x^2 + 4)$, we write

$$\frac{2x^2 - x + 4}{x^3 + 4x} = \frac{2x^2 - x + 4}{x(x^2 + 4)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 4}$$

Multiplying by $x(x^2 + 4)$, we have

$$2x^2 - x + 4 = A(x^2 + 4) + x(Bx + C) = (A + B)x^2 + Cx + 4A$$

Equating coefficients, we obtain

$$A + B = 2, \quad C = -1, \quad 4A = 4 \Rightarrow A = 1, \quad B = 1, \quad C = -1$$

And so

$$\begin{aligned} I &= \int \left(\frac{1}{x} + \frac{x-1}{x^2+4} \right) dx = \int \frac{dx}{x} + \int \frac{x}{x^2+4} dx - \int \frac{dx}{x^2+4} \\ &= \ln|x| + \frac{1}{2} \ln(x^2+4) - \frac{1}{2} \arctan \frac{x}{2} + C \end{aligned}$$

Example 10: Evaluate $I = \int \frac{3x + 2}{x^2 + 2x + 2} dx$

Solution : In order to calculate this integral, the first step is to find two numbers A and B such that the numerator

$$3x + 2 = A(2x + 2) + B$$

where $2x + 2$ is the derivative of the denominator. We obtain $A = \frac{3}{2}$ and $B = -1$, and

$$I = \int \frac{3x + 2}{x^2 + 2x + 2} dx = \frac{3}{2} \int \frac{2x + 2}{x^2 + 2x + 2} dx - \int \frac{1}{x^2 + 2x + 2} dx$$

The first integral is

$$\int \frac{2x + 2}{x^2 + 2x + 2} dx = \ln(x^2 + 2x + 2) + C_1$$

and the second

$$\int \frac{1}{x^2 + 2x + 2} dx = \int \frac{dx}{(x + 1)^2 + 1} = \arctan(x + 1) + C_2$$

Therefore,

$$I = \frac{3}{2} \ln(x^2 + 2x + 2) - \arctan(x + 1) + C$$

Example 11: Evaluate $I = \int \frac{1}{x^4 + 1} dx$

Solution : We have:

$$\begin{aligned}x^4 + 1 &= x^4 + 2x^2 + 1 - 2x^2 = (x^2 + 1)^2 - (\sqrt{2}x)^2 \\&= (x^2 + \sqrt{2}x + 1)(x^2 - \sqrt{2}x + 1)\end{aligned}$$

It follows that

$$\frac{1}{x^4 + 1} = \frac{Ax + B}{x^2 + \sqrt{2}x + 1} + \frac{Cx + D}{x^2 - \sqrt{2}x + 1}$$

and

$$1 = (Ax + B)(x^2 - \sqrt{2}x + 1) + (Cx + D)(x^2 + \sqrt{2}x + 1)$$

Equating coefficients, we obtain the following system to determine A , B , C , and D :

$$\begin{cases} A + C = 0 \\ -\sqrt{2}A + B + \sqrt{2}C + D = 0 \\ A - \sqrt{2}B + C + \sqrt{2}D = 0 \\ B + D = 1 \end{cases} \implies \begin{cases} A = \frac{1}{2\sqrt{2}}, B = \frac{1}{2} \\ C = -\frac{1}{2\sqrt{2}}, D = \frac{1}{2} \end{cases}$$

Therefore

$$\begin{aligned} I &= \frac{1}{2\sqrt{2}} \int \frac{x + \sqrt{2}}{x^2 + \sqrt{2}x + 1} dx - \frac{1}{2\sqrt{2}} \int \frac{x - \sqrt{2}}{x^2 - \sqrt{2}x + 1} dx \\ &= \frac{1}{2\sqrt{2}} I_1 - \frac{1}{2\sqrt{2}} I_2 \end{aligned}$$

$$\begin{aligned}
 I_1 &= \frac{1}{2} \int \frac{2x + 2\sqrt{2}}{x^2 + \sqrt{2}x + 1} dx = \frac{1}{2} \int \frac{2x + \sqrt{2} + \sqrt{2}}{x^2 + \sqrt{2}x + 1} dx \\
 &= \frac{1}{2} \int \frac{2x + \sqrt{2}}{x^2 + \sqrt{2}x + 1} dx + \frac{\sqrt{2}}{2} \int \frac{1}{x^2 + \sqrt{2}x + 1} dx \\
 &= \frac{1}{2} \ln(x^2 + \sqrt{2}x + 1) + \arctan(\sqrt{2}x + 1) + C_1 \\
 I_2 &= \frac{1}{2} \int \frac{2x - 2\sqrt{2}}{x^2 - \sqrt{2}x + 1} dx = \frac{1}{2} \int \frac{2x - \sqrt{2} - \sqrt{2}}{x^2 - \sqrt{2}x + 1} dx \\
 &= \frac{1}{2} \int \frac{2x - \sqrt{2}}{x^2 - \sqrt{2}x + 1} dx - \frac{\sqrt{2}}{2} \int \frac{1}{x^2 - \sqrt{2}x + 1} dx \\
 &= \frac{1}{2} \ln(x^2 - \sqrt{2}x + 1) - \arctan(\sqrt{2}x - 1) + C_2
 \end{aligned}$$

Finally,

$$\begin{aligned} I &= \frac{1}{2\sqrt{2}} \left[\frac{1}{2} \ln(x^2 + \sqrt{2}x + 1) + \arctan(\sqrt{2}x + 1) \right] \\ &\quad - \frac{1}{2\sqrt{2}} \left[\frac{1}{2} \ln(x^2 - \sqrt{2}x + 1) - \arctan(\sqrt{2}x - 1) \right] + C \\ &= \frac{1}{4\sqrt{2}} \ln \frac{x^2 + \sqrt{2}x + 1}{x^2 - \sqrt{2}x + 1} + \frac{1}{2\sqrt{2}} \arctan \frac{\sqrt{2}x}{1 - x^2} + C \end{aligned}$$

Note that in the final result we use the formula

$$\arctan x + \arctan y = \arctan \frac{x + y}{1 - xy}$$

CASE IV: The denominator $Q(x)$ contains a repeated irreducible quadratic factor.

If $Q(x)$ has the factor $(ax^2 + bx + c)^r$, where $b^2 - 4ac < 0$, then the sum

$$\frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \cdots + \frac{A_rx + B_r}{(ax^2 + bx + c)^r}$$

occurs in the partial fraction decomposition of $\frac{R(x)}{Q(x)}$. Each of the terms of this expression can be integrated by using a substitution or by first completing the square if necessary.

Example 12: Evaluate $I = \int \frac{1 - x + 2x^2 - x^3}{x(x^2 + 1)^2} dx$

Solution : The form of the partial fraction decomposition is

$$\frac{1 - x + 2x^2 - x^3}{x(x^2 + 1)^2} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1} + \frac{Dx + E}{(x^2 + 1)^2}$$

Multiplying by $x(x^2 + 1)^2$, we have

$$\begin{aligned} 1 - x + 2x^2 - x^3 &= A(x^2 + 1)^2 + (Bx + C)x(x^2 + 1) \\ &+ x(Dx + E) = A(x^4 + 2x^2 + 1) \\ &+ B(x^4 + x^2) + C(x^3 + x) + Dx^2 + Ex \\ &= (A + B)x^4 + Cx^3 + (2A + B + D)x^2 \\ &+ (C + E)x + A \end{aligned}$$

If we equate coefficients, we get the system

$$A + B = 0, \quad C = -1, \quad 2A + B + D = 2, \quad C + E = -1, \quad A = 1$$

which has the solution $A = 1$, $B = -1$, $C = -1$, $D = 1$, and $E = 0$. Thus

$$\begin{aligned} I &= \int \left(\frac{1}{x} - \frac{x+1}{x^2+1} + \frac{x}{(x^2+1)^2} \right) dx \\ &= \int \frac{dx}{x} - \frac{1}{2} \int \frac{2x dx}{x^2+1} - \int \frac{dx}{x^2+1} + \frac{1}{2} \int \frac{2x dx}{(x^2+1)^2} \\ &= \ln|x| - \frac{1}{2} \ln(x^2+1) - \arctan x - \frac{1}{2} \cdot \frac{1}{x^2+1} + C \end{aligned}$$

INTEGRATION OF TRIGONOMETRIC FUNCTIONS

Strategy for Evaluating $\int \sin^m x \cos^n x \, dx$

- (a) If the power of cosine is odd $n = 2k + 1$, save one cosine factor and use $\cos^2 x = 1 - \sin^2 x$ to express:

$$\int \sin^m x \cos^n x \, dx = \int \sin^m x (1 - \sin^2 x)^k \cos x \, dx$$

Then substitute $u = \sin x$

- (b) If the power of sine is odd $m = 2k + 1$, save one sine factor and use $\sin^2 x = 1 - \cos^2 x$ to express:

$$\int \sin^m x \cos^n x \, dx = \int (1 - \cos^2 x)^k \cos^n x \sin x \, dx$$

Then substitute $u = \cos x$

Strategy for Evaluating $\int \sin^m x \cos^n x \, dx$

- (c) If the powers of both sine and cosine are even, use the half-angle identities:

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

It is sometimes helpful to use the identity

$$\sin 2x = 2 \sin x \cos x$$

Example 13: Find $I = \int \sin^3 x \, dx$

$$\begin{aligned} I &= \int \sin^2 x \cdot \sin x \, dx = \int (1 - \cos^2 x) \sin x \, dx \\ &= \left| u = \cos x \Rightarrow du = -\sin x \, dx \right| = \int (u^2 - 1) du \\ &= \frac{u^3}{3} - u + C = \frac{\cos^3 x}{3} - \cos x + C \end{aligned}$$

Example 14: Find $I = \int \cos^5 x \sin^2 x \, dx$

$$\begin{aligned} I &= \int \cos^4 x \sin^2 x \cos x \, dx = \int (1 - \sin^2 x)^2 \sin^2 x \cos x \, dx \\ &= \left| u = \sin x \Rightarrow du = \cos x \, dx \right| = \int (1 - u^2)^2 u^2 \, du \\ &= \int (u^6 - 2u^4 + u^2) \, du = \frac{u^7}{7} - \frac{2u^5}{5} + \frac{u^3}{3} + C \\ &= \frac{\sin^7 x}{7} - \frac{2\sin^5 x}{5} + \frac{\sin^3 x}{3} + C \end{aligned}$$

Example 15: Find $I = \int \cos^4 x \, dx$

$$\begin{aligned} I &= \int \left(\frac{1 + \cos 2x}{2} \right)^2 dx = \int \left(\frac{1 + 2 \cos 2x + \cos^2 2x}{4} \right) dx \\ &= \int \left(\frac{1 + 2 \cos 2x + \frac{1 + \cos 4x}{2}}{4} \right) dx \\ &= \int \left(\frac{3}{8} + \frac{\cos 2x}{2} + \frac{\cos 4x}{8} \right) dx \\ &= \frac{3}{8}x + \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C \end{aligned}$$

Strategy for Evaluating $\int \frac{\tan^m x}{\cos^n x} dx$

- (a) If the power of cosine is even ($n = 2k$, $k \geq 2$), save a factor of $\frac{1}{\cos^2 x}$ and use $\frac{1}{\cos^2 x} = 1 + \tan^2 x$ to express the remaining factors in terms of $\tan x$. Then substitute $u = \tan x$
- (b) If the power of tangent is odd $m = 2k + 1$, save a factor of $\frac{\tan x}{\cos x}$ and use $\tan^2 x = \frac{1}{\cos^2 x} - 1$ to express the remaining factors in terms of $\frac{1}{\cos x}$. Then substitute $u = \frac{1}{\cos x}$.

Example 16: Find $I = \int \frac{\tan^6 x}{\cos^4 x} dx$

$$\begin{aligned} I &= \int \tan^6 x \frac{1}{\cos^2 x} \frac{1}{\cos^2 x} dx = \int \tan^6 x (1 + \tan^2 x) \frac{dx}{\cos^2 x} \\ &= \left| u = \tan x \Rightarrow du = \frac{dx}{\cos^2 x} \right| = \int u^6 (1 + u^2) du \\ &= \frac{u^7}{7} + \frac{u^9}{9} + C = \frac{\tan^7 x}{7} + \frac{\tan^9 x}{9} + C \end{aligned}$$

Example 17: Find $I = \int \frac{\tan^5 x}{\cos^7 x} dx$

$$\begin{aligned} I &= \int \tan^4 x \cdot \frac{1}{\cos^6 x} \cdot \frac{\tan x}{\cos x} dx \\ &= \int \left(\frac{1}{\cos^2 x} - 1 \right)^2 \left(\frac{1}{\cos x} \right)^6 d \left(\frac{1}{\cos x} \right) \\ &= \int (u^2 - 1)^2 u^6 du = \int (u^{10} - 2u^8 + u^6) du \\ &= \frac{u^{11}}{11} - \frac{2u^9}{9} + \frac{u^7}{7} + C \\ &= \frac{1}{11 \cos^{11} x} - \frac{2}{9 \cos^9 x} + \frac{1}{7 \cos^7 x} + C \end{aligned}$$

Example 18: Find $I = \int \tan^3 x \, dx$

$$\begin{aligned} I &= \int \tan x \cdot \tan^2 x \, dx = \int \tan x \left(\frac{1}{\cos^2 x} - 1 \right) dx \\ &= \int \tan x \frac{dx}{\cos^2 x} - \int \tan x \, dx \\ &= \frac{\tan^2 x}{2} + \ln |\cos x| + C \end{aligned}$$

Example 19: Find $I = \int \frac{dx}{\cos^3 x}$

$$I = \int \frac{1}{\cos x} \cdot \frac{dx}{\cos^2 x} = \left| \begin{array}{l} u = \frac{1}{\cos x}, dv = \frac{dx}{\cos^2 x} \\ \Rightarrow du = \frac{\tan x}{\cos x} dx, v = \tan x \end{array} \right|$$

$$= \frac{\tan x}{\cos x} - \int \frac{\tan^2 x}{\cos x} dx$$

$$= \frac{\tan x}{\cos x} - \int \frac{1}{\cos x} \left(\frac{1}{\cos^2 x} - 1 \right) dx$$

$$= \frac{\tan x}{\cos x} - \int \frac{dx}{\cos^3 x} + \int \frac{dx}{\cos x}$$

$$= \frac{\tan x}{\cos x} - I + \int \frac{dx}{\cos x} = \frac{\tan x}{\cos x} - I + \frac{1}{2} \ln \left| \frac{1 + \sin x}{1 - \sin x} \right|$$

$$\Rightarrow I = \int \frac{dx}{\cos^3 x} = \frac{1}{2} \cdot \frac{\sin x}{\cos^2 x} + \frac{1}{4} \ln \left| \frac{1 + \sin x}{1 - \sin x} \right| + C$$

We have another way to solve this problem.

$$I = \int \frac{dx}{\cos^3 x} = \int \frac{\cos x \, dx}{\cos^4 x} = \int \frac{\cos x \, dx}{(1 - \sin^2 x)^2}$$

Putting $t = \sin x$ we obtain the integral of rational function

$$I = \int \frac{dt}{(1 - t^2)^2} = \int \frac{dt}{(1 - t)^2(1 + t)^2}$$

The function under the integral sign can be factored as follows:

$$\frac{1}{(1 - t)^2(1 + t)^2} = \frac{A}{1 - t} + \frac{B}{(1 - t)^2} + \frac{C}{1 + t} + \frac{D}{(1 + t)^2}$$

where $A = B = C = D = \frac{1}{4}$.

Therefore,

$$\begin{aligned} I &= \frac{1}{4} \left[-\ln |1 - t| + \frac{1}{1 - t} + \ln |1 + t| - \frac{1}{1 + t} \right] + C \\ &= \frac{1}{4} \left[\ln \left| \frac{1 + t}{1 - t} \right| + \frac{2t}{1 - t^2} \right] + C \end{aligned}$$

Now, if we substitute $t = \sin x$ back in the above formula, we obtain the same result:

$$I = \int \frac{dx}{\cos^3 x} = \frac{1}{4} \ln \left| \frac{1 + \sin x}{1 - \sin x} \right| + \frac{1}{2} \cdot \frac{\sin x}{\cos^2 x} + C$$

To evaluate the following integrals we use the corresponding identity:

$$(a) \int \sin(mx) \sin(nx) dx$$

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$(b) \int \cos(mx) \cos(nx) dx$$

$$\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$$

$$(c) \int \sin(mx) \cos(nx) dx$$

$$\sin A \cos B = \frac{1}{2}[\sin(A - B) + \sin(A + B)]$$

Example 20: Evaluate $I = \int \sin 4x \cos 7x \, dx$

$$\begin{aligned} I &= \int \frac{1}{2} [\sin(4x - 7x) + \sin(4x + 7x)] \, dx \\ &= \frac{1}{2} \int [-\sin 3x + \sin 11x] \, dx \\ &= \frac{1}{2} \left(\frac{\cos 3x}{3} - \frac{\cos 11x}{11} \right) + C \end{aligned}$$

Strategy for Evaluating the Integral of General Trigonometric Function

$$\int R(\sin x, \cos x) dx$$

where $R(x, y)$ is a rational function of its variables. In general, we can use the substitution $t = \tan \frac{x}{2}$ to rationalize the function R with respect to t . We have

$$\cos x = \frac{1 - t^2}{1 + t^2}, \quad \sin x = \frac{2t}{1 + t^2}, \quad dx = \frac{2dt}{1 + t^2}$$

Example 21: Evaluate $I = \int \frac{dx}{4 \cos x + 3 \sin x + 5}$

Solution : Let $t = \tan \frac{x}{2}$ and

$$\begin{aligned} I &= \int \frac{1}{4 \frac{1-t^2}{1+t^2} + 3 \frac{2t}{1+t^2} + 5} \cdot \frac{2dt}{1+t^2} \\ &= \int \frac{2dt}{4(1-t^2) + 6t + 5(1+t^2)} = \int \frac{2dt}{t^2 + 6t + 9} \\ &= 2 \int \frac{dt}{(t+3)^2} = -\frac{2}{t+3} + C = -\frac{2}{\tan \frac{x}{2} + 3} + C \end{aligned}$$

INTEGRATION OF SOME IRRATIONAL FUNCTIONS

Strategy for Evaluating

$$\int R \left[x, \left(\frac{ax + b}{cx + d} \right)^{m_1/n_1}, \left(\frac{ax + b}{cx + d} \right)^{m_2/n_2}, \dots \right] dx$$

where $R(x, \dots)$ is a rational function of its variables and $m_1, n_1, m_2, n_2, \dots$ are integers. Let N be a Smallest Common Multiple of $n_1, n_2, \dots$. Put

$$\frac{ax + b}{cx + d} = t^N$$

we obtain an integral of a rational function with respect to t .

Example 22: Find $I = \int \frac{dx}{(\sqrt[4]{x+3}-1)\sqrt{x+3}}$

Solution : The integrand consists of $(x+3)^{1/4}$ and $(x+3)^{1/2}$.
The Smallest Common Multiple of 4 and 2 is $N = 4$. Put

$$x+3 = t^4 \Rightarrow dx = 4t^3 dt, \sqrt[4]{x+3} = t, \sqrt{x+3} = t^2$$

$$\begin{aligned}\Rightarrow I &= \int \frac{4t^3 dt}{(t-1)t^2} = 4 \int \frac{t}{t-1} dt = 4 \int \frac{t-1+1}{t-1} dt \\ &= 4 \int \left(1 + \frac{1}{t-1}\right) dt = 4(t + \ln |t-1|) + C \\ &= 4(\sqrt[4]{x+3} + \ln |\sqrt[4]{x+3} - 1|) + C\end{aligned}$$

Example 23: Find $I = \int \frac{dx}{\sqrt{x} - \sqrt[3]{x}}$

Solution : The integrand consists of $x^{1/2}$ and $x^{1/3}$. The Smallest Common Multiple of 2 and 3 is $N = 6$. Put

$$x = t^6 \Rightarrow dx = 6t^5 dt, \sqrt{x} = t^3, \sqrt[3]{x} = t^2, \sqrt[6]{x} = t$$

$$\begin{aligned}\Rightarrow I &= \int \frac{6t^5 dt}{t^3 - t^2} = \int \frac{6t^3}{t - 1} dt \\&= \int \left(6t^2 + 6t + 6 + \frac{6}{t - 1} \right) dt \\&= 2t^3 + 3t^2 + 6t + 6 \ln |t - 1| + C \\&= 2\sqrt{x} + 3\sqrt[3]{x} + 6\sqrt[6]{x} + 6 \ln |\sqrt[6]{x} - 1| + C\end{aligned}$$

Example 24: Find $I = \int \frac{1}{x} \sqrt{\frac{1-x}{1+x}} dx$

Solution : Putting $t = \sqrt{\frac{1-x}{1+x}}$ gives us $x = \frac{1-t^2}{1+t^2}$, $dx = \frac{-4t}{(1+t^2)^2} dt$, and

$$\begin{aligned} I &= \int \frac{1+t^2}{1-t^2} \cdot t \cdot \frac{-4t}{(1+t^2)^2} dt = \int \frac{-4t^2 dt}{(1-t)(1+t)(1+t^2)} \\ &= \int \left[-\frac{1}{1-t} - \frac{1}{1+t} + \frac{2}{1+t^2} \right] dt \\ &= \ln \left| \frac{1-t}{1+t} \right| + 2 \arctan t + C \\ &= \ln \left| \frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}} \right| + 2 \arctan \sqrt{\frac{1-x}{1+x}} + C \end{aligned}$$

Strategy for Evaluating

$$\int R\left(x, \sqrt{ax^2 + bx + c}\right) dx, \quad (a \neq 0)$$

where $R(x, \dots)$ is a rational function of its variables. The first step is to use the substitution $u = x + \frac{b}{2a}$ and we obtain the following cases:

1. $\int R\left(u, \sqrt{\alpha^2 - u^2}\right) du \Rightarrow u = \alpha \sin t \quad \text{or} \quad u = \alpha \tanh t$
2. $\int R\left(u, \sqrt{\alpha^2 + u^2}\right) du \Rightarrow u = \alpha \tan t \quad \text{or} \quad u = \alpha \sinh t$
3. $\int R\left(u, \sqrt{u^2 - \alpha^2}\right) du \Rightarrow u = \frac{\alpha}{\cos t} \quad \text{or} \quad u = \alpha \cosh t$

Example 25: Find $I = \int \frac{\sqrt{4-x^2}}{x^2} dx$. Let $x = 2 \sin t$, where $-\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$. Then $dx = 2 \cos t dt$ and

$$\sqrt{4-x^2} = \sqrt{4-4\sin^2 t} = \sqrt{4\cos^2 t} = 2|\cos t| = 2\cos t$$

and

$$\begin{aligned} I &= \int \frac{2\cos t}{4\sin^2 t} 2\cos t dt = \int \frac{\cos^2 t}{\sin^2 t} dt = \int \left(\frac{1}{\sin^2 t} - 1 \right) dt \\ &= -\cot t - t + C \end{aligned}$$

We have $\sin t = \frac{x}{2}$, $\cos t = \frac{\sqrt{4-x^2}}{2}$, and $t = \arcsin \frac{x}{2}$.

Then

$$I = -\frac{\sqrt{4-x^2}}{x} - \arcsin \frac{x}{2} + C$$

Example 26: Evaluate $I = \int \sqrt{x^2 - a^2} dx$

Solution : Let $x = a \cosh t \Rightarrow \sqrt{x^2 - a^2} = a \sinh t$, $dx = a \sinh t dt$, and

$$\begin{aligned} I &= a^2 \int \sinh^2 t dt = \frac{a^2}{2} \int (\cosh 2t - 1) dt \\ &= \frac{a^2}{2} \left(\frac{\sinh 2t}{2} - t \right) + C = \frac{a^2}{2} (\sinh t \cosh t - t) + C \\ &= \frac{1}{2} x \sqrt{x^2 - a^2} - \frac{a^2}{2} \ln |x + \sqrt{x^2 - a^2}| + C \end{aligned}$$

Example 27: Evaluate $I = \int x^2 \sqrt{a^2 + x^2} dx \quad (a > 0)$

Solution : Let $x = a \sinh t \Rightarrow \sqrt{a^2 + x^2} = a \cosh t, dx = a \cosh t dt, t = \ln \left(\frac{x + \sqrt{a^2 + x^2}}{a} \right)$ and

$$\begin{aligned} I &= \int (a^2 \sinh^2 t)(a \cosh t)(a \cosh t dt) \\ &= a^4 \int \sinh^2 t \cosh^2 t dt = \frac{a^4}{4} \int \sinh^2(2t) dt \\ &= \frac{a^4}{4} \int \frac{\cosh(4t) - 1}{2} dt = \frac{a^4}{8} \left(\frac{1}{4} \sinh(4t) - t \right) + C_1 \end{aligned}$$

$$\begin{aligned}
 I &= \frac{a^4}{8} \sinh t \cosh t (\sinh^2 t + \cosh^2 t) - \frac{a^4}{8} t + C_1 \\
 &= \frac{a^4}{8} \cdot \frac{x}{a} \cdot \frac{\sqrt{a^2 + x^2}}{a} \left(\frac{x^2}{a^2} + \frac{a^2 + x^2}{a^2} \right) \\
 &\quad - \frac{a^4}{8} \ln \left(\frac{x + \sqrt{a^2 + x^2}}{a} \right) + C_1 \\
 &= \frac{x(2x^2 + a^2)\sqrt{a^2 + x^2}}{8} - \frac{a^4}{8} \ln(x + \sqrt{a^2 + x^2}) + C
 \end{aligned}$$

where $C = C_1 + \frac{a^4}{8} \ln a$.

That's all. Thanks a lot

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